Feature-Resolved Attention: A Clean Derivation

Dmitry Manning-Coe

May 22, 2026

Abstract

Feature-resolved attention (FRA) decomposes an attention circuit in the basis of sparse autoencoder (SAE) features. The goal is to assign a scalar target, such as a downstream feature preactivation or steering direction, to specific query, key, and value-side feature contributions. This note gives a clean derivation of the two pieces we use most often: frozen-attention OV attribution and first-order QK attribution via the softmax Jacobian. It also records the distinction between feature ranking and intervention path, since conflating these two choices has been a recurring source of confusion.

Contents

1 Decomposition

Our goal is to decompose the attention block into linearly independent contributions from each feature. To do so, we must first obtain the decomposition of the residual stream vector $\mathbf{x}$ into a linear combination $\mathbf{u}$ of Sparse Autoencoder features:

$$\mathbf{u} \equiv \sigma\left(W^{\text{enc}}\mathbf{x} + \mathbf{b}^{\text{enc}}\right). \tag{1}$$

We will typically work in component form and so we note that:

$$u_q^\lambda = \sigma\left(\sum_a W_a^{\text{enc},\lambda} x_{qa} + b^{\text{enc},\lambda}\right). \tag{2}$$

Note that in the SAE literature it is conventional to associate each latent feature λ with a decoder direction $\mathbf{f}^\lambda$ in residual-stream space. Under the column-vector convention used here, this is the image of the latent basis vector $\mathbf{e}^\lambda$ under the decoder:

$$\mathbf{f}^\lambda \equiv W^{\text{dec}}\mathbf{e}^\lambda = W^{\text{dec}}_{:,\lambda}. \tag{3}$$

Equivalently, in row-vector SAE implementations where $\mathbf{x}' = \mathbf{u}W^{\text{dec}} + \mathbf{b}^{\text{dec}}$, the same feature direction is the corresponding row $W^{\text{dec}}_{\lambda:}$.

The latent vector $\mathbf{u}$ is then projected back by the decoder matrix to obtain the reconstructed activation:

$$\mathbf{x}' \equiv W^{dec}\mathbf{u} + \mathbf{b}^{dec}. \tag{4}$$

This reconstruction will not, in general, perfectly recover the original activation. The difference defines the reconstruction error $\boldsymbol{\varepsilon}$[1]:

$$\mathbf{x} = \mathbf{x}' + \boldsymbol{\varepsilon}. \tag{5}$$

The component-wise attention update from the resid-pre hookpoint is given by:

$$x^{\text{mid}}_{qa} = x^{\text{pre}}_{qa} + Attn[\mathbf{x}^{\text{ln1}}]_{qa}. \tag{6}$$

To lighten notation, we use the convention that $\mathbf{x}$ with no superscript is the activation vector at the $ln1$ hookpoint. Following [?], we decompose the attention block into an attention pattern A and an OV component:

$$Attn(\mathbf{x})_{qa} = \sum_k A(\mathbf{x})_{qk} OV(\mathbf{x})_{ka}. \tag{7}$$

The attention pattern A is given by the contraction of the Q and K matrices for each head h:

$$A(\mathbf{x})_{qk} = \frac{1}{\sqrt{d_h}} \sum_h \sum_{a,b,i} \left(Q^h_{ia} x_{qa} + b^{Q,h}_i \right) \left(K^h_{ib} x_{kb} + b^{K,h}_i \right), \tag{8}$$

expanding out the terms:

$$A(\mathbf{x})_{qk} = \frac{1}{\sqrt{d_h}} \sum_h \sum_{a,b,i} Q^h_{ia} x_{qa} K^h_{ib} x_{kb} + b^{Q,h}_i K^h_{ib} x_{kb} + Q^h_{ia} x_{qa} b^{K,h}_i + b^{Q,h}_i b^{K,h}_i \tag{9}$$

This is a bit of a mess, so we can probably do better by dropping component form and writing:

$$A(\mathbf{x}) = \mathbf{x}^T Q^T K \mathbf{x} + \underbrace{(\mathbf{b}^Q)^T K}_{\equiv (\mathbf{B}^Q)^T} \mathbf{x} + \mathbf{x}^T \underbrace{Q^T \mathbf{b}^K}_{\equiv \mathbf{B}^K} + \underbrace{(\mathbf{b}^Q)^T \mathbf{b}^K}_{\equiv B^{QK}}. \tag{10}$$

The OV component is given as:

$$OV(\mathbf{x})_{qa} = \sum_h \sum_{a,i} \left(W^{O,h}_{ai} \left(W^{V,h}_{ib} x_{qb} + b^O_i \right) + b^{V,h}_a \right), \tag{11}$$

or, in vector form:

$$OV(\mathbf{x}) = W^O (W^V \mathbf{x} + \mathbf{b}^V) + \mathbf{b}^O = W^O W^V \mathbf{x} + \mathbf{B}^{OV}, \tag{12}$$

where we define:

$$\mathbf{B}^{OV} \equiv W^O \mathbf{b}^V + \mathbf{b}^O. \tag{13}$$

To feature-resolve, we first need to insert the reconstructed activations $\mathbf{x} = \mathbf{x}' + \boldsymbol{\varepsilon}$. For completeness, we keep track of the error term explicitly. The attention pattern becomes:

$$A(\mathbf{x}' + \boldsymbol{\varepsilon}) = A(\mathbf{x}') + A(\boldsymbol{\varepsilon}) + \boldsymbol{\varepsilon}^T Q^T K \mathbf{x}' + (\mathbf{x}')^T Q^T K \boldsymbol{\varepsilon} - B^{QK}, \tag{14}$$

where the last term just comes from overcounting the bias. The OV term is:

$$OV(\mathbf{x}' + \boldsymbol{\varepsilon}) = OV(\mathbf{x}') + OV(\boldsymbol{\varepsilon}) - \mathbf{B}^{OV}. \tag{15}$$

To feature-resolve, we simply propagate through the feature-wise decomposition of the activations. That is, we replace:

$$x'_{qa} \rightarrow X^\lambda_{qa} \equiv W^{dec,\lambda}_a u^\lambda_q + b^{dec}_a, \tag{16}$$

[1] Note that, since we do know the exact activations in all the cases we consider, we can measure the error term and see how it interacts with the features!

through the attention block. We note that the reconstruction error $\boldsymbol{\varepsilon}$ is defined only after reconstruction, and hence cannot be decomposed into underlying features. Putting the feature-resolved activation X into the attention pattern gives:

$$A(\mathbf{x}') = XQ^TKX + (\mathbf{B}^Q)^TX + X^TQ^T\mathbf{B}^K + B^{QK} \tag{17}$$

In component form:

$$A(\mathbf{x}')_{qk} = \sum_{h=1}^{n}\sum_{i=1}^{d_h}\sum_{a,b=1}^{d_M} X^{\mu}_{qa}Q_{ai}K_{ib}X^{\nu}_{bk} + B^Q_a X^{\nu}_{aq} + X^{\mu}_{qa}(B^K)_a + B^{QK}, \tag{18}$$

and similarly for the attention pattern on $\boldsymbol{\varepsilon}$. Doing the same for the OV:

$$OV(\mathbf{x}') = W^{OV}X + \mathbf{B}^{OV}, \tag{19}$$

becomes in component form:

$$OV(\mathbf{x}')^{\lambda}_{qa} = \sum W^{OV}_{ab}X^{\lambda}_{qb} + B^{OV}_a. \tag{20}$$

Putting all of this together, we can decompose the full attention block as:

$$Attn(x)_{qa} = \sum_k A(x' + \varepsilon)_{qk}OV(x' + \varepsilon)_{ka} \tag{21}$$

$$= \sum_k \left[A(\mathbf{x}') + A(\boldsymbol{\varepsilon}) + \boldsymbol{\varepsilon}^TQ^TK\mathbf{x}' + (\mathbf{x}')^TQ^TK\boldsymbol{\varepsilon} - B^{QK}\right]_{qk}\left[OV(\mathbf{x}') + OV(\boldsymbol{\varepsilon}) - \mathbf{B}^{OV}\right]_{ka} \tag{22}$$

We note that models that use Rotary Attention Patterns (RoPE) can be accommodated in this framework by introducing modified position-dependent QK matrices:

$$\tilde{Q}^h_{q,ia} = R^h_{q,ij}Q^h_{ja}, \qquad \tilde{K}^h_{k,ib} = R^h_{k,ij}K^h_{jb}, \tag{23}$$

A Appendix Placeholders